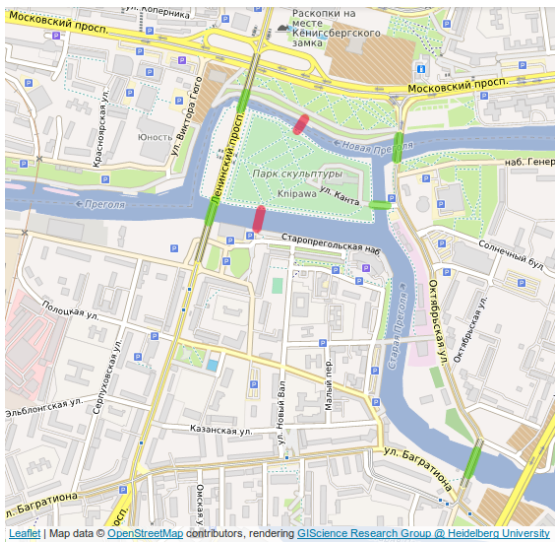
Worksheet #19; date: 10/31/2018
MATH 55 Discrete Mathematics

1. *(Rosen 10.3.51)* A simple graph G is called self-complementary if G and $\bar{G}$ are isomorphic. Find a self-complementary simple graph with five vertices.

2. *(Rosen 10.3.52)* Show that if G is a self-complementary simple graph with v vertices, then $v \equiv 0$ or $1 \pmod 4$.

3. *(Rosen 10.3.55)* How many nonisomorphic simple graphs are there with five vertices and three edges.

4. Modern Königsberg bridge problem: Was there an Euler path before the bombs? Is there an Euler path after the bombs?



5. Suppose a graph has the incidence matrix:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

   More generally, if you know the incidence matrix A, how do you check if the graph has a Euler path? What about Euler circuit?

6. *(Rosen 10.5.28b)* For which values of m and n does the complete bipartite graph $K_{m,n}$ have an Euler path?

7. Show that the Petersen graph has a Hamilton path but no Hamilton circuit.